⚡ Section: Electricity

🔷 Topic: Sources, Internal Resistance

Question 14 – Using a Graph to Analyse a Circuit

A graph of $\frac{1}{I}$ (A^{-1}) against R (Ω) is provided.

We are told this graph is based on the equation derived from conservation of energy in a circuit, and is in the form:

$$\frac{1}{I} = \frac{r}{E}R + \frac{1}{E}$$

This is of the form $y = mx + c$
Where:

- Gradient $m = \frac{r}{E}$
- Intercept $c = \frac{1}{E}$

(a)(i) Determine the internal resistance r of the battery

✔1 mark

From the graph:

- Gradient $m = \frac{\Delta y}{\Delta x} = \frac{14-2}{1.0-0.2} = \frac{12}{0.8} = 15.0$

We know:

$$\frac{r}{E} = 15.0$$

We'll use this in (ii) to find r.

(a)(ii) Determine the EMF E of the battery

✔2 marks

From the y-intercept:

$$\frac{1}{E} = 2 \Rightarrow E = \frac{1}{2} = 0.50\,\mathrm{V}$$

Now from above:

$$\frac{r}{0.50} = 15.0 \Rightarrow r = 7.5\,\Omega$$

(a)(iii) Determine the short-circuit current

✔2 marks

Short circuit: $R = 0 \Rightarrow I = \frac{E}{r}$

$$I = \frac{0.50}{7.5} = 0.067\,\mathrm{A}$$

(b) Effect of adding a second variable resistor in parallel

✔2 marks

✅ **Answer:**

> The short-circuit current will be **the same**
> because it depends **only on the EMF and internal resistance**, and **not on the external resistance**.

🧠 Revision Tips

- This question uses **graph interpretation** to extract circuit quantities
- **Intercept** gives $\frac{1}{E}$, **gradient** gives $\frac{r}{E}$
- **Short circuit current** = $\frac{E}{r}$, when $R = 0$
- Adding parallel resistance affects total load, but **not** short-circuit curr